The Complete Personal Biography of Phana Phley (KON)

Developer, Independent Builder, and Creator of the PK Ecosystem

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Document Nature:	Personal Archive, Portfolio & Autobiography	**Lifecycle Status:**	Living Document - Actively Building

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1. Who I Am

My legal name is **Phley Phana**, following the Cambodian naming tradition where the family name (Phley) precedes the given name (Phana). In digital software communities, open-source projects, and technical discussions, I am known as **Phana Kurosaki**, and my most frequent handle and nickname is **KON**.

I am an independent software builder, web developer, and technology explorer from **Cambodia**. Everything I build is tied together under my personal brand mark: **PK**, representing **Phana Kurosaki**.

I am not an enterprise executive or an elite corporate specialist. I am an everyday person driven by genuine curiosity for computing, a deep love for software craftsmanship, and an unwavering commitment to building useful, reliable, and aesthetically refined products for my society in Cambodia and the wider world.

On GitHub, I maintain two complementary accounts:

- **@not-Phana:** My primary repository home, where public releases, documentation, and live deployments are hosted.
- **@Kon-Kurosaki:** My lead developer identity, directly authoring hundreds of commits, architecture refactors, and core engine code.

My work spans web applications, developer tools, vector graphics systems, audio engineering, and interactive simulation engines. Above all, I define myself by what I actually construct: starting from a blank directory, writing code line by line, confronting bugs head-on, and turning rough ideas into working software.

2. My Biggest Dream

My biggest dream is to become a fully capable, highly skilled, and completely independent software engineer and product creator—someone who can architect, design, write, deploy, and maintain software systems that solve real problems, starting with Cambodia and expanding outward.

I want to build an ecosystem of software that embodies three core virtues:

1. **Utility:** Every application must serve an unmistakable, practical purpose rather than acting as a vanity showcase.
2. **Speed and Efficiency:** Software should be featherweight, respecting user hardware, launching instantly, and remaining responsive even on entry-level machines.
3. **Dignity and Respect for Users:** No deceptive telemetry, no intrusive advertising, no artificial subscription walls for basic functionality, and no bloated dependencies.

In Cambodia, technology education and specialized digital tools often lag behind. Khmer script typography, Unicode input ergonomics, and localized learning resources are historically fragmented. My dream is to prove that world-class, beautifully engineered software can be created right here in Cambodia by an independent builder who refuses to compromise on quality.

3. Why I Use AI

I use Artificial Intelligence not as a shortcut to bypass hard work, but as a dynamic intellectual amplifier, a tireless technical educator, and an always-available pair programmer.

Coming from a self-taught background in Cambodia, I did not have access to senior software architects or expensive computer science labs. When encountering obscure system behaviors—such as the way Windows injects artificial Control keydown events when AltGr is pressed, or how the Web Audio API coordinates BiquadFilterNode sweeps—there is rarely a human tutor available at 2:00 AM to explain the underlying mechanics.

AI bridges this mentorship divide. I use AI to:

- **Demystify Complex Architecture:** Break down Monaco Editor text models, Abstract Syntax Tree (AST) parsing, Server-Sent Events (SSE) streaming protocols, and Electron IPC boundaries.
- **Accelerate the Feedback Loop:** Receive rapid diagnostic feedback when an algorithm misbehaves, tracing edge cases and discovering where assumptions broke down.
- **Explore Alternative Implementations:** Compare approaches (e.g., HTML5 Canvas direct path rendering vs. DOM-based sprite translation) to understand memory tradeoffs and render performance.
- **Draft and Refine Boilerplate:** Generate baseline structural boilerplate so I can dedicate my cognitive energy to core logic, edge-case handling, and user experience.

4. My Philosophy: AI Should Make Me More Capable, Not Dependent

> *AI must make me more capable as a thinker and engineer; it must never render me dependent or helpless.*

I reject the culture of "vibe coding"—blindly prompting an AI, copying uninspected code, pasting it into a project, and hoping for the best. When something breaks, a vibe coder cannot explain why, cannot trace the call stack, and cannot fix the bug without begging the model for another patch.

For me, every interaction with an AI model is an educational contract:

- If an AI suggests a regular expression for Khmer Unicode syllable clustering, I study the Unicode code blocks (`\u1780-\u17D3`) and verify why each character class exists.
- If an AI helps structure an Electron preload script, I examine the `contextBridge`, study why `nodeIntegration: false` and `contextIsolation: true` prevent remote code execution, and verify the attack vectors myself.
- If an AI generates a draft module, I refactor it, strip unnecessary code, align it with my strict naming conventions, and verify its runtime behavior in real browsers.

My foundational standard is simple: **If every AI service on Earth shut down tomorrow, the knowledge I acquired while building must remain inside my head.** AI is the bicycle for my mind; I am the one steering, pedaling, and navigating the terrain.

5. My Journey into Programming

My software journey began in early 2026, starting with the fundamental building blocks of the web: HTML5, CSS3, and vanilla JavaScript.

- **January 2026 — First Public Milestone: Phana Coffee Shop (@not-Phana):** Established my primary GitHub presence and published my first multi-page web application (*Phana-cofee-shop*), deployed live to Vercel. Grounded myself in semantic HTML, CSS flexbox/grid, DOM filtering algorithms, multi-page routing, and production deployment pipelines.
- **August 2026 — Desktop Systems and Native APIs: Arachne Clock & PK Music:** Expanded into Windows desktop engineering with Electron, React, and TypeScript. Built *Arachne Desktop Clock* (persistent alarms, lap timer, system tray integration) and *PK Music v0.1* (offline Windows music player with local folder traversal, ID3/FLAC parsing, custom `pkmedia://` protocol, and 10-band Web Audio BiquadFilterNode equalizer).
- **September 2026 — Developer Tooling & Core Architecture: PK Studio Code & KON:** Explored developer environment internals. Prototyped *PK Studio Code* around Monaco Editor with multi-tab state, AST diagnostics, plugin bus, and local SSE server. Concurrently created *KON*, an AI coding companion workbench utilizing Server-Sent Events streaming.
- **September 2026 — Flagship Releases: PK Khmer Type & PK Icons Studio:** Shipped *PK Khmer Type* to solve touch-typing education for Khmer Standard, Khmer NiDA, and English QWERTY layouts. Published *PK Icons Studio*, an interactive vector customizer with 50 original 24x24 stroke SVG icons.
- **October 2026 — Engine Expansion & Game Development: v2.4 Studio & NEON PROTOCOL:** Scaled *PK Khmer Type* to version 2.4 Pro Studio (40 levels, 218 lessons, 664 exercises, 3D kinematic dual-hand finger guides, live telemetry dashboard). Built *NEON PROTOCOL // OVERDRIVE* (60 FPS HTML5 Canvas action game) and *Neon Core Defense* (Roblox cooperative wave defense in Luau with Rojo).

6. My Learning Goals and Preferred Learning Style

How I Learn Best:

- **Building Over Watching:** Passive video tutorials do not create understanding. True learning occurs when opening an empty editor, initializing a Git repository, hitting console errors, and solving them.
- **First-Principles Deconstruction:** Taking complex tools apart into foundational components. Building an icon library to master SVG coordinates, or building an audio player to understand Web Audio filter nodes.
- **Granular, Concrete Explanations:** Seeking precise explanations of memory flow, data structures, and state transitions rather than vague analogies.
- **Deliberate Mistake Tracing:** Treating every error as an instructor by inspecting stack traces, adding diagnostic logs, and exploring DOM event phases.

Active Learning Targets: Khmer Unicode normalization and cluster rendering; React 18/19 internals and Vite bundling; Language Server Protocols (LSP) and AST parsing; Rust and WebAssembly (Wasm) for high-performance native utilities.

7. My Projects and the PK Ecosystem

Project Name	Category	Status	Key Technologies	Repository / Location
PK Khmer Type	Educational / Web	Live · Active	Vanilla ES6+, CSS3, Web Audio API, Three.js, LocalStorage	github.com/not-Phana/PK-Khmer-Type
NEON PROTOCOL // OVERDRIVE	Action Game	Live on Pages	HTML5 Canvas 2D, Web Audio API, Vanilla JS, Custom SVG	github.com/not-Phana/PK-game
PK Icons Studio	Developer Tools / UI	Live · Expanding	SVG Vector Geometry, JavaScript, HTML5, CSS3	github.com/not-Phana/PK-icons
PK Studio Code	Developer IDE	In Development	Monaco Editor, Node.js SSE Server, Gemini API	Local Workspace Prototype
KON (PK Intelligence)	AI Workbench	Experiment	Node.js, Express, Server-Sent Events, REST APIs	Local Workspace Prototype
Phana Coffee Shop	Web Application	Published	HTML5, CSS3, JavaScript DOM Filtering	github.com/not-Phana/Phana-cofee-shop
PK Music (v0.1)	Desktop Audio	Prototype	Electron, Web Audio API (10-Band EQ), Node.js	E:\Code\pk-music-electron
Arachne Clock	Desktop Utility	Prototype	Electron, React, TypeScript, Vite	E:\Code\clock
Neon Core Defense	Co-op Sci-Fi Game	Functional	Roblox Luau, Rojo, Raycasting, Procedural Map	E:\Code\make game in roblox

8. PK Khmer Type and Its Purpose

PK Khmer Type (v2.4 Pro Studio Edition) is my flagship public educational project. In Cambodia, typing Khmer is exceptionally challenging because the abugida script (33 consonants, 23 dependent vowels, 12 independent vowels, diacritics, and subscript Coeng signs) spans across four modifier layers (Base, Shift, Ctrl, AltGr).

I engineered a standalone, zero-dependency browser application designed to operate seamlessly both online and 100% offline from the local filesystem (`file:///` protocol). It features:

- Instant switching between **Khmer Standard**, **Khmer NiDA**, and **English QWERTY**.
- A 40-level progressive curriculum with 218 lessons and 664 exercises featuring strict progressive character unlocking.
- Dual-hand 3D kinematic finger guides (44% default opacity) dynamically pointing to target keys and modifier holds.
- Procedural Web Audio switch synthesizer (Topre, Holy Panda, Cherry MX Blue, Linear) with zero external audio assets.
- Live Mastery Dashboard (`dashboard.html`) with per-key latency/accuracy heatmaps and JSON profile backup/restore.
- Zero-Lag Performance Mode to disable all animations for pure mechanical input speed.
- Full bilingual UI (English / Khmer) toggled with a single shortcut (`Alt + L`).

9. PK Studio Code

An in-development developer environment exploring how code editors function internally. Built around the Microsoft Monaco Editor, it integrates multi-tab model states, bracket colorization, code minimap, an interactive terminal REPL, a real-time AST problem diagnostic panel, an extensible lifecycle plugin bus (`PKPluginSystem`), five custom dark themes, and a local Node.js Server-Sent Events live server.

10. KON, My AI Assistant Concept

Named after my nickname KON, this project is a local AI developer workbench. Instead of detached conversational chat, KON is anchored to code development: built with a secure Express backend proxy, progressive SSE token streaming, and targeted diagnostic modes (Explain Code, Debug & Fix, Refactor Architecture, Generate Tests) with multi-provider routing (Gemini and OpenRouter).

11. PK Icons Studio

An open-source collection of 50 pixel-perfect 24x24 stroke SVG vector icons with an interactive browser customizer. Born directly from my strict **Zero-Emoji Policy**, it allows real-time stroke weight, size, and color adjustments, live tag searching, one-click raw SVG markup copying, and batch ZIP archive exports.

12. PK Intelligence and My Future Vision for PK OS

PK Intelligence is my umbrella concept for ambient, non-intrusive software intelligence integrated directly into developer workflows. *PK OS* is a conceptual long-term research dream—an exploration of what a developer-centric operating environment would look like if built with zero bloat, instant boot times, keyboard-first ergonomics, local privacy, and zero unsolicited cloud telemetry.

13. My Relationship with AI Coding Agents, Especially Google Antigravity

I draw a clear line between conversational chatbots (web interfaces providing text summaries) and autonomous coding agents like **Google Antigravity** (Google DeepMind). Antigravity runs shell commands, reads and writes files across project trees, inspects live viewports via Chrome DevTools Protocol (CDP), and verifies builds autonomously. Working with Antigravity represents true pair programming: the human architect directs the vision, philosophy, and taste, while the autonomous agent refactors, tests, and verifies code in real time.

14. My Approach to Software Design and UI/UX

Guided by classical Greek principles: **Poiesis** (Build into existence), **Episteme** (Deep understanding of craft), and **Melete** (Disciplined iterative refinement).

- **Restraint (Meden Agan — Nothing in Excess):** Every element must serve a distinct purpose. No clutter, no decorative filler.
- **Zero-Emoji Rule:** Clean, crisp monoline SVG icons exclusively. No inconsistent or childish emojis.
- **Dark-Mode Precision:** Obsidian bases, charcoal surfaces, gold/bronze accent borders, and subtle cyan status indicators.
- **Zero-Lag Performance Mode:** Animations must never block users; provide toggles to disable all visual effects for instant responsiveness.
- **Bilingual Ergonomics:** Full first-class support for both Khmer script typography and English.

15. My Computer Hardware and Technical Limitations

Workstation Specifications: **Intel Xeon W-1370P CPU @ 3.60GHz (8 cores, 16 threads)**, **32.0 GB RAM**, **Integrated Intel UHD Graphics P750 (no discrete GPU)**, Windows 11 Pro, and SSD workspace on Drive E:\.

Operating without a dedicated GPU prevents running massive local 70B LLMs or heavy local diffusion pipelines, but I treat this constraint as an engineering advantage. It forces me to write featherweight, zero-dependency JavaScript and 60 FPS Canvas loops that run smoothly on modest laptops across Cambodia.

16. My Experiences with AI Subscriptions, Usage Limits, and Affordable Tools

Independent builders must navigate subscription costs and rate limits carefully. I focus on cost-effective, high-reasoning models (Google Gemini Flash, OpenRouter), utilize free-tier developer API quotas, design token-efficient prompts, and build local tools (KON, PK Studio Code) to avoid vendor lock-in.

17. Challenges I Encounter While Building Software

Major hurdles include: complex Khmer Unicode syllable normalization and Windows AltGr event deduplication; managing 60 FPS Canvas draw calls and Web Audio lifecycles; solitary debugging without senior mentors; and auditing AI hallucinations to ensure architectural accuracy.

18. My Personality, Humor, and Interests

Self-aware, honest, resilient, and witty. My workspace includes folders like "*i dont freaking know im doing*" alongside production repositories—a genuine reflection of the learning process. I am passionate about technology, loyal to my community, and committed to continuous improvement.

19. My Interest in Bleach, Roblox, and Technology

Bleach: The alias *Phana Kurosaki* and nickname *KON* are direct tributes to Tite Kubo's Bleach (Ichigo Kurosaki and Kon the lion). The series' themes of perseverance, protection, and sharp aesthetic contrast inspire my work.
Roblox: Building *Neon Core Defense* in Luau using Rojo taught me networked multiplayer replication, authoritative game loops, and raycast aiming.
Technology: Passionate about studying browser engines (Brave Core), media frameworks (VLC), and low-level computing architectures.

20. My Approach to Learning Through Building Real Projects

```
[Curiosity / Problem]
        |
        v
[Blank Directory Initialization]
        |
        v
[Initial Architecture & Prototype]
        |
        v
[Encounter Real Errors & Edge Cases]
        |
        v
[Research Specs, Docs & AI Consultation]
        |
        v
[Iterative Refactoring & Optimization]
        |
        v
[Shipped, Working Product]
```

Every major skill I possess—DOM event handling, Web Audio BiquadFilterNodes, Windows AltGr event interception, Canvas game loops—was mastered by building real software from scratch.

21. My Beliefs About AI, Creativity, Programming, and Independence

AI does not eliminate creativity; it removes mechanical friction so human vision and architectural taste can shine. Programming is structured logical reasoning, and the human engineer remains entirely responsible for correctness and architecture. True independence means understanding systems deeply enough to build with or without AI.

22. My Future Ambitions

1. Expand *PK Khmer Type* into the national educational touch-typing standard across schools and universities in Cambodia.
2. Refactor and publish *PK Studio Code* to the global open-source community as an accessible, featherweight IDE.
3. Expand *PK Icons Studio* into an extensive, multi-category zero-emoji SVG resource.
4. Master low-level systems programming in Rust and C++ to build blazing-fast native tools.
5. Inspire and mentor other young Cambodian builders to create impactful software.

23. First-Person Autobiography: An Honest Introduction

Hello. My name is **Phley Phana**. Most people online know me as **Phana Kurosaki**, and my friends and fellow builders call me **KON**.

I was born and live in Cambodia, a country with an ancient cultural heritage and a young, rapidly emerging digital future. I didn't begin with private tutors or a high-end gaming PC. I began with curiosity, an internet connection, an integrated-graphics computer, and a desire to see my ideas exist in the real world.

When I first opened a code editor in early 2026, I started like everyone else: making simple web pages with HTML and CSS, testing colors, and figuring out why a layout broke when the browser resized. My first public milestone on GitHub was *Phana Coffee Shop*, deployed to Vercel. It proved to me that if I put in the hours, code could become something people could actually visit and use.

From that moment on, I wanted to understand how deeper software worked. I experimented with Electron to build native Windows desktop tools: an offline audio player with a real 10-band equalizer (*PK Music*) and a desktop alarm utility (*Arachne Clock*). I explored game loops by writing a 60 FPS HTML5 Canvas action game (*NEON PROTOCOL // OVERDRIVE*) and building cooperative wave defense games in Luau on Roblox (*Neon Core Defense*).

Then I looked at the software environment around me in Cambodia. Typing in our native Khmer language has always been a steep hurdle across four keyboard layers. I built *PK Khmer Type*—a complete, zero-dependency typing studio featuring 40 levels, 218 lessons, dual-hand 3D kinematic finger guides, procedural audio switch synthesis, and a live mastery telemetry dashboard that works 100% offline.

To support my projects, I established the **PK** brand. Across all my work, I follow a strict **Zero-Emoji Policy**, crafting crisp 24x24 vector icons, which gave birth to *PK Icons Studio*. I began exploring IDE internals and AI companions through *PK Studio Code* and *KON*.

Throughout this journey, I have used Artificial Intelligence extensively. Tools like Google Antigravity, Claude, ChatGPT, and Codex have been my digital mentors, sounding boards, and co-authors. But I hold one rule above all others: **AI is here to make me a stronger engineer, not a helpless passenger.** I never accept code I don't understand, and I continuously study the mechanics behind the screen.

I am an independent builder at the beginning of a long journey. I have built real things, I have made countless mistakes, and I have learned from every single one of them. My goal is simple: to keep building software that is fast, useful, honest, and empowering.

24. The Most Important Things I Want People to Understand About Me

1. **I Am an Active Builder, Not a Theorist:** I measure myself by working software. I initialize repositories, write code, debug failures, and ship tools people can use.
2. **I Build for Real People and My Society:** Projects like *PK Khmer Type* exist because I care deeply about helping people in Cambodia master technology and communicate fluently.
3. **I Value Craft, Restraint, and Zero Bloat:** I eliminate unnecessary animations, avoid invasive tracking, ban childish emojis in favor of crisp SVG vectors, and engineer applications to run smoothly on modest hardware.
4. **I Control the Tools; the Tools Do Not Control Me:** I embrace AI enthusiastically as an accelerator, but I insist on understanding every line of code in my projects. If the tools vanish, my knowledge

remains.

5. **My Journey Is Just Beginning:** Every project I build is a stepping stone toward becoming a world-class independent engineer.

Document compiled and archived on Drive E: (Phana Kurosaki Workstation) · Phnom Penh / Cambodia · October 2026
Repository Ecosystem: https://github.com/not-Phana · https://github.com/Kon-Kurosaki